Math 432 Implementation Homework 7

Due date: 14 June Sunday

You can get help from AI but write down your own code

1. (10 points) Write a program that computes the Linear Approximation Table (LAT) of a given DES S-box. The input is a DES S-box and the output is its LAT.
2. (20 points) Construct a 5-round linear characteristic for DES in which the round function is active in every round. Compute the theoretical bias of the characteristic using the Piling-up Lemma. Then estimate the bias experimentally by encrypting a sufficiently large number of plaintexts and comparing the observed bias with the theoretical value. Determine approximately how many plaintext-ciphertext pairs are required before the experimental estimate becomes close to the theoretical bias.
3. (20 points) Using the 5-round linear characteristic obtained in Question 2, perform a 1-round linear cryptanalysis attack against 6-round DES. Recover the sixth-round subkey bits involved in the attacked S-boxes and verify experimentally that the correct subkey can be distinguished from wrong guesses.